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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
However, because of the fixed-shape segmentation, all morphological RFs were excluded.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: , women: 266 age: 51.2 +- 11.4 years) was used in this study .
Randomization
In accordance with the instruction manual, the 9 simulated nodules were randomly attached to the internal structure of the lung by using double-sided adhesive tape and tweezers.
Blinding
not detected.
Power Analysis
not detected.
Replication
For the intra-CT protocol trial, Aquilion ONE TSX-301C was selected to obtain CT images with varying acquisition and reconstruction parameters (Table 2), and the intra-CT protocol trial was repeated twice to obtain a more general result.
Number: For the intra-CT protocol trial, Aquilion ONE TSX-301C was selected to obtain CT images with varying acquisition and reconstruction parameters (Table 2), and the intra-CT protocol trial was repeated twice to obtain a more general result.

Cell Line Authentication	
Authentication: For further validation of the results of stability analyses , the entire dataset was divided into 2 parts with nonoverlapping CT scanners (see Table S2	
Data Information	
Identifiers: The varying scope of acquisition and reconstruction parameters were chosen based on previous studies 16,18,19,28,29 and radiologists previous knowledge.	16,18,19,28,29

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
PyRadiomics (version 2.2.0) with default calculation settings, 30 an open-source library implemented in Python capable of extracting a large panel of features from medical images, was used to extract the RFs.	Python		Suggestion: python-bx (RRID:SCR_024202)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
TABLE 1 CT Acquisition Protocols for the Test-Retest and Inter-CT Trials Site CT Scanner (Manufacturer) Pitch* Rotation Time (s) Tube Voltage (kVp) Tube Current (mA*s) FOV (mm) Slice Thickness (mm)* Slice Interval (mm)* Reconstruction Kernel+ Reconstruction Method++ Shanghai Public Health Clinical Center (Jinshan District) Aquilion ONE TSX-301C (Toshiba) 0.813 0.5 120 75 350 1.0 1.0 FC56 AIDR 3D (standard) Shanghai Public Health Clinical Center (Jinshan District) SCENARIA (Hitachi) 0.8281 0.5 120 75 350 1.0 1.0 66 Intelli IP (Lv.2) Shanghai Public Health Clinical Center (Hongkou District) Brilliance 64 (Philips) 0.891 0.5 120 75 350 1.0 1.0 L Standard (enhancement = 1.0) Zhongshan Hospital of Fudan University SOMATOM Definition AS (Siemens Healthineers) 0.9 0.5 120 75 350 1.0 1.0 B60f ADMIRE (strength = 3) Zhongshan Hospital of Fudan University Aquilion ONE TSX-301A (Toshiba) 0.828 0.5 120 75 350 1.0 1.0 FC56 AIDR 3D (standard) Zhongshan Hospital of Fudan University UCT550 (United Imaging Healthcare) 0.8875 0.5 120 75 350 1.0 1.0 SHARPC Adaptive filter function (enhancement = 2.5) Shanghai Sixth Peoples Hospital Revolution CT (GE Healthcare) 0.992 0.5 120 75 350 1.25 1.25 LUNG ASiR (PlusSS40) The number of decimal places was consistent with the corresponding CT scanner control panel.	Test-Retest
The Friedman test and pairwise Wilcoxon signed-rank test were used to compare the magnitude of the influence of different acquisition and reconstruction parameters.	Friedman test
	Wilcoxon signed-rank test
The Spearman rank correlation coefficient was used to analyze the consistency of the influencing patterns of different sources of variability.	Spearman rank correlation
For clinical validation, the Wilcoxon rank-sum test was used to explore the discriminative effectiveness of the representative RFs.	Wilcoxon rank-sum test

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